

Grounded Force Conditioning for Behavior-Cloned VLA Policies:

A Drop-In, Retrofittable FiLM Pathway That Brakes Beyond the Demonstrations

Anonymous Authors

Paper ID: XXXX

Abstract—Vision-language-action (VLA) policies increasingly receive wrist wrench and contact signals in their state, yet a behavior-cloned policy given the full wrench can still fail to bind its behavior to it. On a real-robot suction case-picking task where the cases carry battery cells—a 15 N contact-force limit, and a stop-on-contact transition observable only through force—a naively fine-tuned VLA presses through contact in every trial and never stops on its own. We present a conditioning pathway that changes how force is delivered, not what is delivered: a condition vector $\hat{c}$ (contact, force magnitude, seal) computed from signals the policy already receives, injected by FiLM at the state token, with the raw wrench masked. The pathway has three properties. It is *drop-in*: no backbone change, no new information, no cost in imitation error, and it attaches to a second backbone (π_0). It is *retrofittable*: added to an already-trained naive policy, a few thousand steps of continued training suffice. And it *extrapolates in the right direction*: because $\hat{c}$ is a calibrated monotone scalar, the policy brakes harder the harder it is pressed—stopping and retreating at forces beyond the demonstrations, where the naive policy’s response fades—and picks succeed at a stack height absent from training. A matched ablation shows both ingredients are necessary: unmasked, the pathway trains causally dead (*bypass*); ungrounded, it is the worst policy in the study. We also ran it on our robot: every conditioned pick stops on its own (median contact force 2.5 N) where the naive policy is externally aborted in every attempt, and on-robot counterfactual probes reproduce the offline crossover.

Index Terms—imitation learning, vision-language-action models, force control, causal confusion, contact-rich manipulation

I. INTRODUCTION

Behavior-cloned vision-language-action (VLA) policies are rapidly acquiring physical condition signals. Wrist force/torque, contact events, and task-state bits now routinely appear in the observation vector, and a growing line of work fuses force into the policy backbone for contact-rich manipulation [1]–[4]. All of this work shares a validation recipe—add the signal, retrain, and validate by task success (occasionally alongside rollout force statistics [2], [3])—and none of it measures whether the trained policy *causally uses* the signal when it matters: whether, holding everything else fixed, changing the force input changes the action. Imitation learning offers no such guarantee. The objective asks the policy to reproduce *what* the expert did, not *why*: a clone

can imitate the behavior convincingly while being driven by entirely different cues than the expert was. Imitation pins down the trajectory, not the mechanism—equally faithful clones can act from opposite causes, and the loss cannot tell them apart [5], [6].

The consequence is easiest to see where force is the only signal that can carry the task. We study real-robot suction case picking with a single head-mounted camera and no wrist camera: the arm must stop descending the moment the suction cup meets the case, and stack heights differing by a few millimeters—the entire success margin—are visually near-indistinguishable at head-camera range. The wrench is in the state the policy receives, and every demonstration exhibits the stop. Yet a naively fine-tuned VLA—full wrench in its state, hereafter the *naive policy*—presses through contact in every evaluation trial and never initiates a stop of its own: each run ends only when an external safety layer, separate from the policy, trips at its 15 N force limit—set to protect the battery cells the cases carry (15.4–18.7 N measured at abort)—the policy still commanding descent. The task turns on a single transition—stop when force rises—and it is precisely this transition that the policy fails to bind to force.

Our fix changes how force is delivered, not what is delivered. A condition vector $\hat{c}$ (contact, normal force, seal) is *computed* from signals the naive policy already receives and injected through FiLM modulation [7]; the raw wrench is masked from the state so that the same signal does not enter twice, making $\hat{c}$ the *only* pathway from force to action. We call this pathway *grounded* because its content is computed from measured signals, and the policy trained behind it the *conditioned policy* (“FiLM (ours)” in tables and figures). By construction it receives no information the naive policy lacks, and because the pathway is explicit and unique, forcing $\hat{c}$ at inference directly measures its *causal authority*—the fraction of commanded descent the condition can cancel.

The mask is load-bearing. Leave the raw wrench in the state and the conditioning pathway trains causally dead at unchanged validation loss: training routes force through the already-functional raw path and starves the dedicated one. We call this *bypass* and exhibit it in Sec. V; it is why a fix of

this kind cannot be certified after the fact, and why ours holds by construction. The demonstrations complete the recipe: each press-retreat episode presses to a randomized force target and retreats before any seal event, so the force at which the expert stops varies from episode to episode—in the data, depth, appearance, and timing no longer predict the stop; only the force reading does. The same interface that fixes the policy is the instrument that audits it (Sec. IV).

The resulting policy is not uniformly better—it is different exactly where the task is decided. In a matched comparison—same data, same initialization, imitation error indistinguishable—the naive policy *does* respond to force at the demonstrated magnitudes, as strongly as the conditioned policy or more so. But its response is a *template*: sized to the demonstrated forces, it fades or saturates as force rises past them, never strengthening into a stop. The conditioned policy’s response instead grows with force, to the point of stopping and backing away at forces beyond anything in the training data (Sec. V). The mechanism carries to the real closed loop: while the naive policy drives contact force past the 15 N safety limit on every attempt, the conditioned policy stops on its own in every pick—across three stack heights, at a median contact force of 2.5 N—and live counterfactual probes reproduce the offline crossover on the physical system (Sec. V-D).

Our contribution is the pathway and three properties of it, each measured by intervention on the trained policy (Sec. IV-C):

- **Drop-in.** $\hat{c}$ is computed from signals the policy already receives and enters through an identity-initialized FiLM layer at the state token; the backbone is untouched, no information is added, imitation error is unchanged, and the same pathway attaches to a second backbone (π_0), where it again turns a policy that never stops into one that does (Sec. IV, V-C).
- **Retrofittable.** Added to an already-trained naive policy and continued for a few thousand steps on the same demonstrations, the pathway turns a saturating force template into a monotone brake; a matched ablation shows the mask and the grounding are both necessary—unmasked, the pathway trains causally dead (*bypass*); ungrounded, it is the worst policy in the study (Sec. V).
- **Monotone beyond the demonstrations.** Because $\hat{c}$ is a calibrated monotone scalar, braking grows with force past anything demonstrated, stopping and retreating where the naive policy’s template saturates and presses through (Sec. V). The same holds on our robot: every conditioned pick stops under the 15 N abort limit (median contact force 2.5 N), including at a stack height absent from the demonstrations, and live counterfactuals reproduce the crossover (Sec. V-D).

In one sentence: *a policy that stops when it feels what the demonstrations felt and a policy that stops harder the harder it is pressed are indistinguishable in loss and similar at the demonstrated forces—they differ exactly where the task is decided, and only demonstrations in which force alone predicts*

the stop, behind a grounded, masked FiLM pathway, yield the second.

II. RELATED WORK

Force- and tactile-conditioned VLA policies. ForceVLA fuses wrist force as dedicated tokens through a mixture-of-experts backbone [1]; TA-VLA injects torque adapters into the action decoder [8]; PhaForce schedules policy phases on contact and force signals [2]; ForceFlow [3], FARM [9], and FILIC [4] condition diffusion or impedance-loop policies on force and tactile input; TacForeSight learns force-conditioned tactile prediction [10]. These works *design for* force use and validate it with task success and retraining ablations, sometimes alongside rollout force statistics [2], [3]. None of these measurements establishes causal use at the decision moment: an ablation compares two differently-trained functions, rollout statistics are observational, and success can be carried by correlates of force rather than force itself. The nearest analyses stop short of the force input: TA-VLA perturbs input tokens of its *baseline* to site the adapters and reports correlational representation similarity [8]; Tactile-VLA’s dose-response intervenes on the *instruction*, not the force stream [11]; ForceFlow masks its trained policy’s *visual* input and reads out task success [3]. We do not claim these systems bypass their force inputs—their tasks may contain force information that vision cannot substitute, leaving no shortcut to take. Our claim is narrower and falsifiable: *when* the demonstrations correlate the condition with cheaper cues, a policy with a dedicated force pathway can train to indistinguishable loss while routing around that pathway (we exhibit exactly this with our unmasked control), and no metric in the standard validation stack detects it. We contribute the missing measurement: an inference-time intervention on the trained policy’s force input, with the action as the readout.

Intermediate condition representations. Injecting a compact, task-level representation between observation and action is a recurring design: RT-Affordance conditions on predicted affordance plans [12], and FiLM modulation [7] is the standard mechanism for conditioning a backbone on side information. Our $\hat{c}$ shares the shape but differs in role: it is *computed* from signals the policy already receives—not a learned head that could itself be bypassed—and it is paired with a mask that removes the raw signal, turning the representation from a convenience into the policy’s sole, measurable route to force.

Causal confusion and shortcut learning. That imitators latch onto spurious correlates is well established: causal confusion in IL [5], copycat behavior on observation histories [13], and shortcut learning in deep networks broadly [6], [14]. Recent work documents shortcuts in generalist robot policies [15] and uses targeted interventions to expose or correct them [16], [17]. The symptoms are on record in force-conditioned policies themselves: FuSe observes fine-tuned generalists ignoring newly added sensors [18], and FM-VLA diagnoses a history-length shortcut in its own force stream and patches it with augmentation—verifying the fix, again, only by task success [19]. We instantiate this literature

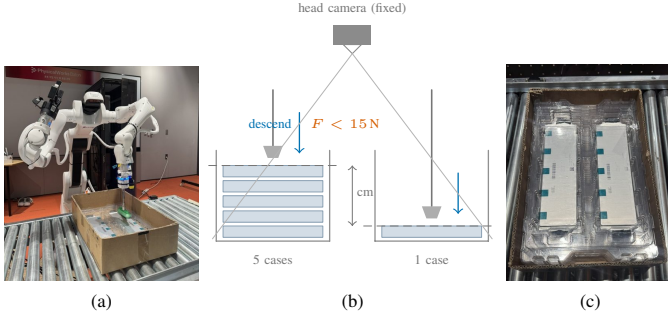


Fig. 1. Setting. (a) The dual-arm mobile manipulator picking a battery-cell case from a tray with a suction cup. (b) The stack height varies across episodes, so the contact plane moves by centimeters, while the head camera keeps the same pose; a few millimeters of over-travel take contact force past the 15 N limit. (c) The head-camera view of a tray: from this range the remaining stack height is visually near-indistinguishable across layers, so the stop-on-contact transition must be carried by force, not vision.

where it has not been applied—force conditioning in fine-tuned VLAs—and add the usage/form dissociation: the data’s correlation structure decides *whether* a signal is used, while the conditioning pathway decides *what form* its use takes under extrapolation.

Privileged supervision and interventional evaluation. Privileged-teacher distillation [20], [21] assumes access to a signal worth distilling; our probes answer the prior question of whether the deployed policy uses the signal at all. The closest interventional precedent is ForceSight, which ablates predicted force *goals* at its controller under matched initial conditions [22]; the intervention sits on the output side of a model that takes no force input, and reads out end-of-episode success rather than the action. Our press-retreat prescription is a data-side intervention in the spirit of decorrelation fixes for causal confusion [5], but designed from a physical argument (Sec. VI) that reactive gating is arithmetically insufficient on stiff contact, so the demonstrations must teach anticipatory approach and retreat.

III. TASK, SETTING, AND THE FAILURE

Task and platform. We study suction case picking on a Dexmate Vega-1p, a dual-arm mobile manipulator: the arm descends onto a stack of rigid cases, must stop when the suction cup meets the top case, hold until vacuum seal is confirmed, and lift (Fig. 1a). The cases carry battery cells, which tolerate little compression—over-pressing risks the product, not just the pick—so deployment enforces a 15 N contact-force limit. The margin is tight in both directions: too little press and the seal fails; a few millimeters too much and contact force climbs past the limit, since the stack is stiff (~ 5 N/mm, Sec. VI). The policy controls a 7-D end-effector action (3-D position and 3-D rotation increments plus a suction command) at 15 Hz, executed in chunks of 5 actions (~ 330 ms between observations).

Observations. The policy observes a single head-mounted camera (RGB and colorized depth; no wrist camera) and a 15-D state: end-effector position and quaternion, suction

command, vacuum-seal bit, and the 6-D wrist wrench. The setting is a deliberately clean diagnostic: at head-camera range, case-stack heights differing by millimeters—the entire success margin—are visually near-indistinguishable (Fig. 1b,c), while the wrench and seal bit signal the contact transition directly. Every condition the task depends on is *in the state the policy receives*; the question is whether the policy binds its behavior to it.

Demonstrations. We collect 100 training episodes (19,445 frames) and hold out 6 validation episodes (1,169 frames) with a scripted expert designed to *decorrelate* force from its habitual correlates: each episode descends with suction on, presses to a randomized force target drawn from $[8, 15]$ N, retreats 5–10 mm from the contact plane, then hovers and lifts immediately on seal. Episodes are collected at two different stack heights, so the contact plane itself moves across the dataset. Every episode thus demonstrates the counterfactual the task requires: force rises $\rightarrow$ stop $\rightarrow$ move back up, *before* any seal event, at a force level that varies independently of depth. This is the data-side prescription whose rationale we develop in Sec. VI; here it matters that the naive policy and all conditioned variants are trained on exactly these episodes.

The failure. Deployed on the robot, the naive policy—a VLA fine-tuned on these demonstrations, with the full wrench and seal bit in its state (backbone in Sec. V)—fails every evaluation trial the same way: it descends, makes contact, and keeps pressing—never initiating a stop of its own—until the external safety layer aborts the run (15.4–18.7 N measured at abort). The transition that the demonstrations exhibit in every single episode, driven by the one signal that observably announces it, is the transition the policy does not perform. The rollout says the policy is bad; it does not say *why*. The policy might ignore force entirely; it might respond to force but too weakly; it might respond in a form that collapses precisely when force exceeds the demonstrated range. These hypotheses have opposite prescriptions (more data? different sensors? different architecture?), and none of them is distinguishable from training curves or rollouts. Distinguishing them requires intervening on the signal itself—the instrument we build next.

IV. THE CONDITIONING PATHWAY

The pathway has two parts: a conditioning route built so that force reaches the action in one explicit, calibrated form (IV-A–IV-B), and a battery of counterfactual probes that measure how much that route moves the action (IV-C). The route is designed to be *drop-in* (no backbone change, no new information), *retrofitable* (attachable to a trained policy), and *measurable*: because it is explicit and unique, every component makes one number well-defined—how much does the condition causally move the action?

A. Computed condition vector $\hat{c}$

From the wrench F and seal bit already present in the naive policy’s state we compute a small set of task-critical

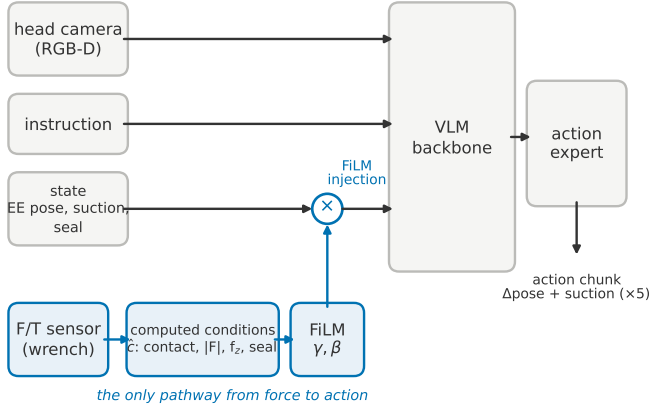


Fig. 2. The conditioned policy. The raw wrench does not enter the policy state; force reaches the action only as the computed condition vector $\hat{c}$, injected through FiLM modulation of the state token entering the VLM backbone. Because this pathway is explicit and unique, forcing $\hat{c}$ at inference measures the causal authority of the condition.

conditions—the physical cues a human operator references (“if you feel contact, stop; if the seal is on, lift”):

$$\begin{aligned}\hat{c}_{\text{contact}} &= \text{clip}((\|F\| - F_0) / \tau, 0, 1), \\ \hat{c}_{\text{fmag}} &= (\|F\| - F_0) / \tau_m \quad (\text{unsaturated}), \\ \hat{c}_{f_z} &= (f_z - o_z) / \tau_z, \quad \hat{c}_{\text{seal}} = \text{seal},\end{aligned}\quad (1)$$

with offsets and scales calibrated to the measured force statistics of the demonstrations (e.g., F_0 at the contact-onset magnitude).¹ Two properties are essential. *Computed, not learned*: $\hat{c}$ is a fixed function of observed signals, so its physical meaning cannot drift during training—which is what makes forcing $\hat{c}$ at inference a well-defined intervention rather than a perturbation of an arbitrary latent. *Zero new information*: every channel is a deterministic function of state dimensions the naive policy also receives, so conditioning changes *routing*, never *content*—the “it works because it got extra sensing” objection is excluded by construction. The channels are deliberately plural: keeping contact, force magnitude, and seal separate is what later lets the probes attribute authority to specific signals (Sec. V-A).

B. FiLM injection and the force mask

$\hat{c}$ enters the policy through FiLM modulation [7] at the state token entering the VLM backbone; the injection site matters, and Sec. V-C compares it against the action token. The FiLM layer is zero-initialized so that it starts as the identity, and the backbone is untouched: at the moment the pathway is attached, the conditioned policy *is* the already-trained policy, and can be fine-tuned in place from there. Every offline conditioned policy in this paper is obtained this way, by continuing the trained naive checkpoint (Sec. V). The policy still receives the

¹ Calibration quality matters and is itself a finding. The conditioned policies in this paper are separately trained instances of the same design: the robot picks (Sec. V-D) and the replication pair quoted in Sec. V-B use an earlier calibration; the recipe ablation and the live counterfactuals (Sec. V-E) use offsets re-estimated from the measured contact distribution plus the unsaturated $\hat{c}_{\text{fmag}}$ channel.

force information—only the structure of its delivery changes. Because $\hat{c}$ is computed *from* the wrench, leaving the raw wrench in the state would hand the policy the same signal twice, once in each form; we therefore mask the wrench dimensions from the state (Fig. 2). Forcing $\hat{c}$ at inference then manipulates *all* of the force information the policy receives: if the policy responds to force at all, it must show in how the action changes with $\hat{c}$ —the quantity the probes below measure.

C. Evaluation: counterfactual probe battery

All probes intervene on a trained, frozen policy; none involves retraining. The offline probes (P1–P4) run on held-out validation episodes. We define **authority** as the intervention-induced change in the commanded vertical action on committed-descent frames, expressed as a percentage of the mean commanded descent: 0% means the intervention is ignored, 100% means it fully cancels descent, >100% means reversal (retreat).

(P1) Condition forcing (open loop). How much does the condition alone move the action? With images and state frozen, $\hat{c}$ is forced from its observed value to a target pattern—either all channels saturated (*full forcing*) or a calibrated contact-onset pattern matching measured first-contact statistics (*contact-moment forcing*). Applicable to conditioned models only.

(P2) State transplant (open loop). Does the policy respond to a *real* contact state, and through which route? The six wrench dimensions of a measured first-contact state (the median wrench at the first force rise, pooled over episodes; the seal bit keeps its pre-contact value) are transplanted into pre-contact descent frames. This probe applies to any policy, including the naive policy—it is what puts both policies on the same scale. Routing the transplant selectively, to the $\hat{c}$ computation alone (d_{FiLM}) or the raw state alone (d_{raw}), attributes the response to each route.

(P3) Dose-response sweep. Does the policy brake *more* the harder it is pressed? A policy that has bound braking to force should; one that has memorized a contact template should respond most near the template and fade beyond it. The transplanted wrench is rescaled to 8, 10, and 12 N and the response traced as a function of force *dose*; since the demonstrations peak near the low end of this range, the upper doses measure extrapolation. The sweep’s *slope sign* is the paper’s central discriminator.

(P4) Closed-loop press simulation. Does the policy actually stop before the force limit when no seal ever comes? Open-loop probes cannot show consequences, so we place the frozen policy in a closed loop at each episode’s nominal first-contact timing: commanded descent advances a virtual penetration depth, a linear-stiffness force model raises the simulated wrench accordingly, and the seal event is withheld—the operationally critical misalignment scenario in which no press depth ever produces a seal. Outcomes: whether the policy stops before the force limit, and maximum penetration. A control variant grants the seal at a shallow depth, separating force-driven stopping from seal-event-driven stopping.

(P5) Live counterfactual probe. Do the offline responses hold on the real robot, where the sensor, the contact, and the control loop are real rather than replayed or simulated? We freeze observations at staged poses along the approach and apply (P1)–(P3) interventions to *two* policies—the conditioned policy and the naive policy—on identical frozen observations, so both receive the same physical counterfactual.

Scope. P1–P3 are single-frame and open-loop; P4 closes the loop under a force model; P5 closes the gap to the real system. No single probe is decisive—the findings below rest on their agreement.

V. EXPERIMENTS

The backbone throughout is SmolVLA [23], a compact open VLA chosen so that a matched ablation—four policies from one initialization—and the continued-training runs are affordable; Sec. V-C repeats the central result on the 3B-class π_0 . The offline experiments compare four policies that start from the *same* trained naive checkpoint and continue training on the *same* demonstrations: the naive policy itself, the full recipe (the conditioned policy), and two controls that each remove one ingredient: one keeps the raw wrench in the state alongside $\hat{c}$ (no mask), the other keeps the mask but trains on $\hat{c}$ values shuffled across the batch, so $\hat{c}$ no longer tracks the measured force (no grounding) (Sec. V-B). Imitation error on held-out episodes is indistinguishable across all four (0.84–0.96 mm per step): on the data manifold, these are the same policy. Every reported model is its validation-loss-selected (*best*) checkpoint—one uniform selection rule for all policies. This is the retrofit setting of Sec. IV-B: continued training ran for 20k steps, validation loss selects the conditioned policy’s checkpoint at 2,500 steps, and the monotone form reported below persists to the final checkpoint (+0.97 $\rightarrow$ +3.44 mm at 8 $\rightarrow$ 12 N).

A. The two policies under intervention

Table I (upper rows) and Fig. 3 give P1–P4 for the naive and conditioned policies; the controls follow in Sec. V-B.

(P1) Condition forcing. Forcing every channel of $\hat{c}$ to saturation moves the conditioned policy’s committed descent by +0.9 mm per frame. Restricting the forcing to a calibrated contact-onset pattern shows the limit of what any policy adopts: even the best-conditioned policy moves by only 7%—the 1–2-frame contact *transient* is weakly adopted across the board.

(P2) State transplant: both policies on one scale. Transplanting a measured first-contact state into pre-contact descent frames moves the naive policy by +1.63 mm—105% of its descent command. The naive policy is not force-blind: its raw route responds strongly to force-bearing state. In the conditioned policy the same transplant lands entirely on d_{FiLM} (+1.57 mm, 94%; the masked raw route contributes nothing)—the mask working as designed. *What* is transplanted matters as much as whether: a *sealed* state moves both models far more than the contact state does (on-robot: 147% vs. 67% for the naive policy; Sec. V-E). Together with the weak

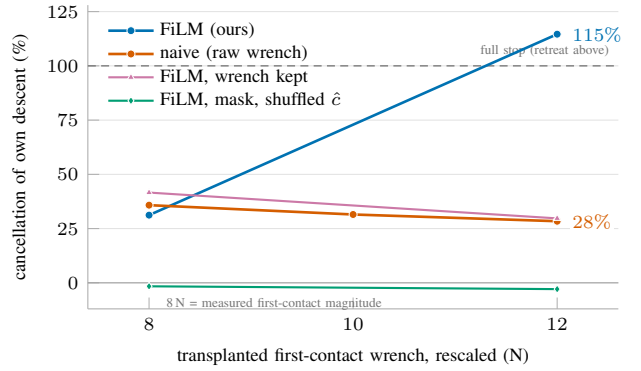


Fig. 3. Dose-response under forced wrench magnitude (P3): cancellation of each policy’s own commanded descent as the transplanted first-contact wrench is rescaled to 8/10/12 N. The naive policy’s response fades past the demonstrated range; the unmasked variant lies on top of it (all response through the raw route); the ungrounded pathway is flat at zero; only the grounded pathway grows with dose, crossing full cancellation into commanded retreat at 12 N.

contact-onset forcing above, the pattern is that stable, persistent conditions (seal, sustained press) are adopted essentially for free, while a transient that is redundant with depth-like cues in ordinary demonstrations is not; Sec. VI draws the consequence for what mechanism can stop a stiff press.

(P3) Dose-response: raw access binds a template, the grounded pathway binds a brake. The naive policy responds to transplanted force—at the demonstrated magnitude. As the dose rises past the demonstrations, its response *fades* (36% $\rightarrow$ 28%): a template, matched ever more poorly as reality departs from it. The conditioned policy’s response *grows* with dose, crossing full cancellation into commanded retreat at 12 N (31% $\rightarrow$ 115%)—where the unsaturated channel reads five times its training range. The monotone extrapolation is not learned from data (no demonstration presses at these forces); it is inherited from the channel’s design as a calibrated monotone scalar. The forms are not artifacts of checkpoint selection: no naive checkpoint we examined exceeds 65% cancellation at any dose, while the grounded brake remains monotone to the end of training. That the naive policy responds *more* at the demonstrated dose does not help it: over-press is the regime in which force rises past the template, so its brake weakens exactly as it becomes necessary, and its strongest response—to the sealed state (Sec. V-E)—is bound to a signal that never arrives in an over-press.

(P4) Press simulation: consequences. Closing the loop under the seal-never press turns the response forms into outcomes. Every conditioned episode settles by 9.8 mm of penetration, with contact force at or below 7.1 N when the simulation ends. The naive policy settles in four of six episodes but runs away in two—still descending when the step budget ends, at up to 14.9 mm with force climbing through 13.1 N; its later checkpoints respond *more* in open loop yet fare *worse* here (3/6, max 16.0 mm), because a response that saturates below full cancellation cannot stop a stiff press regardless of its size. The seal-granted control variant localizes the

TABLE I

OFFLINE BATTERY. UPPER ROWS: THE NAIVE POLICY VS. THE CONDITIONED POLICY (FiLM, OURS). LOWER ROWS: ABLATION CONTROLS WITH THE SAME INITIALIZATION AND DATA AND ONE INGREDIENT REMOVED. ENTRIES ARE THE CHANGE IN COMMANDED DESCENT CAUSED BY THE INTERVENTION; PERCENTAGES ARE RELATIVE TO THE POLICY’S OWN DESCENT ON THE EVALUATED FRAMES (% OWN; NONE FOR P1, WHOSE FRAMES DIFFER). PRESS SIM = THE SEAL-NEVER PRESS SIMULATION. THE NAIVE POLICY HAS NO $\hat{c}$ TO FORCE (—).

Policy	Force access	Forcing (P1) mm/frame	Transplant (P2) mm/frame (% own)	Dose-response (P3), 8 → 12 N mm/frame (% own)	Press sim (P4) max depth (mm)
naive VLA	raw wrench	—	+1.63 (105%)	+1.41 → +1.12 (36→28%) ↓	14.9
FiLM (ours)	masked, grounded $\hat{c}$	+0.9	+1.57 (94%)	+1.20 → +4.40 (31→115%) ↑	9.8
FiLM, wrench kept	raw + dead $\hat{c}$	0.0	+1.68 (111%)	+1.57 → +1.12 (42→30%) ↓	11.9
FiLM, mask, shuffled $\hat{c}$	ungrounded pathway	−0.2	−0.06 (0%)	−0.06 → −0.11 (≈0%)	57.2

failure: given a seal event at shallow depth the naive policy stops normally (0.5 mm over-travel)—withhold the seal and it presses through. Its stopping is real but bound to the *seal event*; the conditioned policy’s is bound to *force*.

B. Ablating the recipe: the mask and the grounding

The two controls run the same battery (Table I, lower rows); each isolates one ingredient.

Remove the mask, and force reroutes: bypass. The unmasked control keeps the raw wrench in the state alongside $\hat{c}$, and behaves as the naive policy everywhere: its dose-response curve lies on top of the naive policy’s (+1.57 → +1.12 vs. +1.41 → +1.12; identical at 12 N), forcing its $\hat{c}$ moves nothing (0.0 mm), and the pathway decomposition places its transplant response entirely on the raw route ($d_{\text{FiLM}}=0.04$). With a redundant raw route available, gradient descent satisfies the imitation objective through the already-functional path and starves the dedicated pathway. This is the *bypass* of Sec. I—and nothing in training registers it: the unmasked variant’s validation loss matches the conditioned policy’s to the fourth decimal (0.14750 vs. 0.14762). An independently trained pair under the earlier calibration reproduces the result: 7% condition-forcing authority with the wrench kept versus 76% with it masked, again at near-identical loss.

Remove the grounding, and the pathway is worse than raw access. The shuffled control has the same added parameters, the same continued training, the same pathway—and zero response at every dose (−0.2 mm under forcing; flat dose-response), with the worst closed-loop outcome in the study: no episode settles, all six still descending at the step budget, 19–23 N past the force limit, up to 57 mm deep. Grounding, not capacity, is the active ingredient: the mask removed the raw route, and the noise-trained $\hat{c}$ replaced it with nothing.

C. The pathway attaches to a second backbone

The same pathway—same channels, same calibration, same mask—attaches to π_0 [24] (3B-class, $\sim 8\times$ the parameter count) at its dense state token, with the backbone otherwise unchanged. Fine-tuned naively on the same demonstrations, π_0 reproduces the naive failure: in the seal-never press no episode settles (mean penetration 17.0 mm, max 19.6). With the pathway attached, five of six settle (mean 7.2 mm) at equal imitation error (0.93 vs. 1.03 mm per step), and the transplant response again runs entirely through the FiLM route (masked

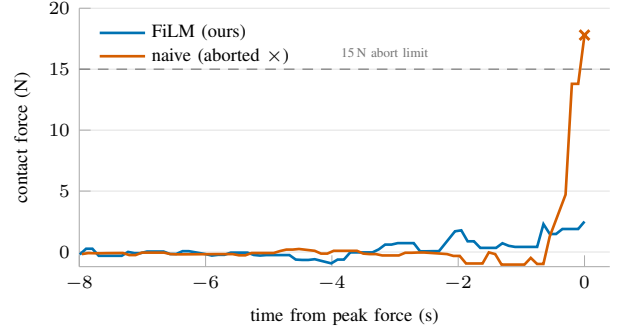


Fig. 4. Contact force on the robot: one representative trial per policy, aligned at peak force ($t=0$), over the interaction window (before lift, where carrying the case would re-enter payload weight into the measured force). The naive attempt rises through the demonstrated range and is terminated by the external 15 N abort ($\times$; all three naive attempts end this way, at 15.4–18.7 N), while the conditioned pick presses to 2.5 N—the median of its nine picks across three stack heights, one of them absent from the demonstrations—and stops on its own.

raw route +0.00 mm). Attaching the same pathway at the *action* token instead stops no episode (22.9 mm mean): the injection site is part of the recipe, not a detail.²

D. Real-robot experiments: closed-loop picks

Beyond the offline battery, we also ran both policies on the robot of Sec. III. The offline results make three predictions about the real closed loop: a template response that fades beyond the demonstrations should let contact force run away until something outside the policy intervenes; a monotone brake should stop on its own, at forces near the demonstrated range; and the two responses should cross between the demonstrated magnitudes and the safety limit. The outcome variable is the *force behavior* of each attempt—task success follows from it by definition. The deployed conditioned policy is an instance of the same design trained from the base model; the retrofitted instance of Sec. V was evaluated offline and in the live probes of Sec. V-E, not in closed loop.

The predictions hold (Fig. 4). Under the naive policy every attempt ends the same way: contact, continued descent, and

²This π_0 instance carries the saturating contact channel but not the unsaturated $\hat{c}_{\text{mag}}$, so its response is threshold-like rather than dose-monotone, and one episode runs to 18.3 mm. Offline only; π_0 was not deployed on the robot.

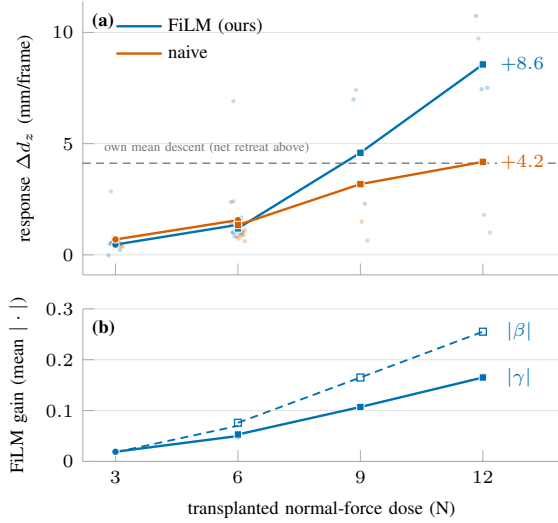


Fig. 5. On-robot counterfactuals (P5): identical frozen observations, wrench transplanted to a fixed normal-force dose (circles: ten-pose session; squares: three-pose session; faint dots: individual poses). (a) The response ordering reverses between 6 and 9 N—the offline crossover on the physical system. (b) FiLM modulation magnitude rises monotonically with dose.

an external force-limit abort at 15.4–18.7 N—the closed-loop realization of a response that saturates below full cancellation. The conditioned policy stops itself in *every* pick, across three stack heights: 3/3 at the naive policy’s own height, and 3/3 at an intermediate height absent from the demonstrations, which contain only the other two. Peak force, measured over the *interaction window* (descent through retreat, before lifting re-enters payload weight into the wrench; the abort monitors the same quantity), stays at or below 3.1 N in seven of the nine picks (median 2.5 N; two picks sealed with no measurable compression at all). The two force distributions do not overlap, and the task outcomes are their threshold readout: 0/3 versus 9/9. The two harder presses (10.2 and 14.5 N) are the brake visible near its limit—both stopped short of the abort threshold and completed the pick where the naive policy, at the same forces, was still commanding descent.

The stopping runs through the conditioning pathway. The deployed policy’s own dose-response is monotone: transplanted wrenches at 8/10/12 N cancel 3% $\rightarrow$ 7% $\rightarrow$ 13% of committed descent—rising with dose, and carried entirely by the FiLM path (the masked raw route contributes +0.00 mm, the mask working as designed). During the picks themselves the modulation is measurably active at the transitions ($|\gamma|$ +48%, $|\beta|$ +79% at contact/seal relative to descent), and forcing $\hat{c}$ at ten frozen poses on the robot moves the commanded action in the correct direction at every staged condition (contact +0.73, sealed +2.09, +6 N +2.18 mm/frame).

E. Real-robot experiments: on-robot counterfactuals

The live probe (P5) freezes observations at staged poses and doses the naive policy and the conditioned policy with identical counterfactual states. Two sessions matter (Fig. 5).

Fair dose. At transplant doses matching the demonstrations ($\leq$ contact magnitude), the *naive* policy responds as strongly or stronger than the conditioned policy (first-contact: 67% vs. 32% of own descent; sealed state: 147% vs. 58%)—in quantitative agreement with the offline dose-response curves at the same doses, where the template peaks (and, as Sec. V-A argues, this does not help it). The conditioned policy’s response is dose-monotone across the staged conditions (hover +0.29 $\rightarrow$ pre-seal +0.87 $\rightarrow$ sealed +1.41 $\rightarrow$ +6 N +1.70 mm/frame).

High dose. Raising the forced normal force to +6/ +9/ +12 N reproduces the offline crossover on the robot: the conditioned policy accelerates its braking, +1.19 $\rightarrow$ +4.59 $\rightarrow$ +8.56 mm/frame—at 12 N, twice its own descent command, i.e., commanded retreat—while the naive policy’s response collapses at the margin (+1.34 $\rightarrow$ +3.18 $\rightarrow$ +4.18; increments +1.84 $\rightarrow$ +1.00). The reversal sits between 6 and 9 N. At near-surface poses the separation is starkest: naive +1.0–1.8 vs. conditioned +7.5 mm/frame at 12 N. Similar at the demonstrated doses, categorically different where the task is decided—on the robot as offline.

Three validity notes. Force-torque baselines were re-anchored at a pre-descent hover before each probe session, and absolute transplant targets were drift-shifted so both policies received the same physical dose; live conditioned-policy magnitudes are conservative lower bounds, as some frozen states are slightly off-manifold for its input distribution. The naive policy’s closed-loop attempts were run at a single stack height—its failure mode is uniform and force-driven, but the height-generalization evidence applies to the conditioned policy only. High-dose probes cover three poses; we report them as targeted supplementary measurements, not a survey.

VI. DISCUSSION AND LIMITATIONS

Why the prescription must be anticipatory. A first-principles calculation explains both the failure and the fix. The case stack is stiff: ~ 5 N/mm of penetration. Stopping below the 15 N limit therefore requires a full stop within ~ 2 mm of contact—under 330 ms action chunks and the demonstrated descent rates, this is arithmetically impossible for *reactive* gating that begins braking at the contact transient, no matter how much authority the transient carries. The viable mechanisms are anticipatory: approach slowly, press deliberately, retreat. The press-retreat demonstrations teach exactly this behavior prior, and the probes confirm the resulting division of labor: the successful policy’s contact-moment authority is small (7%) while its full-condition authority is large (76%)—success comes from an anticipatory behavior prior plus post-hoc state gating, not from a reflex.

A computed condition is an operational knob. Because $\hat{c}$ is computed rather than learned, the pathway creates knobs the naive policy lacks. Force-torque baselines drift (~ 1 N within a day, pose-sensitive); the conditioned policy re-anchors its offsets at a hover pose each run—a deployment-time correction with no retraining—while the baseline’s force statistics are frozen in its normalization.

Limitations. Single task, single platform, small robot- n : the robot results are a case study, and the statistical claims live in the offline battery ($n=6$ held-out episodes $\times$ dense frames, seven policies across two backbones). The retrofitted instance and the π_0 instance were not deployed in closed loop, and the press-retreat demonstrations are a precondition we do not ablate here: the data decides *whether* force is used, the pathway decides the *form*. The press simulation depends on a linear force model; we mitigate with the seal-granted control and the live probes, which bracket the simulated mechanism from both sides. Contact-moment (reactive) authority plateaus low in every configuration we trained—in development phases, oversampling transition frames also failed to move it; distilling an explicit descend-until-contact oracle (e.g., via DAgger [20], [25]) is the natural next step, as is a negative-control task whose critical condition is *not* observable in the state, and replication of the bypass audit on force-fusion architectures whose designs assume use [1], [3].

VII. CONCLUSION

On a task whose critical transition is observable only through force, a behavior-cloned VLA policy given the full wrench pressed through contact in every trial. We changed how force is delivered, not what is delivered: a condition vector computed from signals the policy already receives, injected by FiLM at the state token, with the raw wrench masked. The pathway is drop-in, attaches to an already-trained policy in a few thousand steps, carries to a second backbone, and—because its channels are calibrated monotone scalars—brakes harder the harder it is pressed, stopping and retreating beyond the demonstrated forces and succeeding at an undemonstrated stack height. The same interface makes the policy auditable: forcing the condition measures its causal authority, exposes bypass when the mask is removed, and shows that grounding, not capacity, is the active ingredient. Imitation pins down the trajectory; the pathway pins down the form in which the condition is used—and lets that form be shaped, measured, and corrected.

REFERENCES

- [1] J. Yu *et al.*, “ForceVLA: Enhancing VLA models with a force-aware MoE for contact-rich manipulation,” *arXiv preprint arXiv:2505.22159*, 2025.
- [2] M. Wang *et al.*, “PhaForce: Phase-scheduled visual-force policy learning with slow planning and fast correction for contact-rich manipulation,” *arXiv preprint arXiv:2603.08342*, 2026.
- [3] S. Zhang *et al.*, “ForceFlow: Learning to feel and act via contact-driven flow matching,” *arXiv preprint arXiv:2605.11048*, 2026.
- [4] H. Ge *et al.*, “FILIC: Dual-loop force-guided imitation learning with impedance torque control for contact-rich manipulation tasks,” *arXiv preprint arXiv:2509.17053*, 2025.
- [5] P. de Haan, D. Jayaraman, and S. Levine, “Causal confusion in imitation learning,” in *NeurIPS*, 2019, arXiv:1905.11979.
- [6] R. Geirhos *et al.*, “Shortcut learning in deep neural networks,” *Nature Machine Intelligence*, vol. 2, pp. 665–673, 2020.
- [7] E. Perez, F. Strub, H. de Vries, V. Dumoulin, and A. Courville, “FiLM: Visual reasoning with a general conditioning layer,” in *AAAI*, 2018.
- [8] Z. Zhang *et al.*, “TA-VLA: Elucidating the design space of torque-aware vision-language-action models,” in *CoRL*, 2025, arXiv:2509.07962.
- [9] E. Helmut, N. Funk, T. Schneider, C. de Farias, and J. Peters, “Tactile-conditioned diffusion policy for force-aware robotic manipulation,” *arXiv preprint arXiv:2510.13324*, 2025.
- [10] Y. Zang *et al.*, “TacForeSight: Force-guided tactile world model for contact-rich manipulation,” *arXiv preprint arXiv:2606.11184*, 2026.
- [11] J. Huang, S. Wang, F. Lin, Y. Hu, C. Wen, and Y. Gao, “Tactile-VLA: Unlocking vision-language-action model’s physical knowledge for tactile generalization,” *arXiv preprint arXiv:2507.09160*, 2025.
- [12] S. Nasiriany *et al.*, “RT-Affordance: Affordances are versatile intermediate representations for robot manipulation,” *arXiv preprint arXiv:2411.02704*, 2024.
- [13] C. Wen, J. Lin, T. Darrell, D. Jayaraman, and Y. Gao, “Fighting copycat agents in behavioral cloning from observation histories,” in *NeurIPS*, 2020, arXiv:2010.14876.
- [14] K. Hermann, H. Mobahi, T. Fel, and M. C. Mozer, “On the foundations of shortcut learning,” *arXiv preprint arXiv:2310.16228*, 2023.
- [15] Y. Xing, X. Luo, J. Xie, L. Gao, H. Shen, and J. Song, “Shortcut learning in generalist robot policies: The role of dataset diversity and fragmentation,” *arXiv preprint arXiv:2508.06426*, 2025.
- [16] Y. Chen, Y. Zhang, G. D’urso, N. Lawrance, and B. Tidd, “Improving generalization ability of robotic imitation learning by resolving causal confusion in observations,” *arXiv preprint arXiv:2507.22380*, 2025.
- [17] S. Pfrommer, Y. Bai, H. Lee, and S. Sojoudi, “Initial state interventions for deconfounded imitation learning,” *arXiv preprint arXiv:2307.15980*, 2023.
- [18] J. Jones, O. Mees, C. Sferrazza, K. Stachowicz, P. Abbeel, and S. Levine, “Beyond sight: Finetuning generalist robot policies with heterogeneous sensors via language grounding,” in *ICRA*, 2025, arXiv:2501.04693.
- [19] R. Li *et al.*, “FM-VLA: Force-based memory for vision-language-action models in contact-rich manipulation,” *arXiv preprint arXiv:2607.18231*, 2026.
- [20] D. Chen, B. Zhou, V. Koltun, and P. Krähenbühl, “Learning by cheating,” in *CoRL*, 2019.
- [21] X. Mao, Y. Xu, Z. Sun, E. Miller, D. Layeghi, and M. Mistry, “Learning long-horizon robot manipulation skills via privileged action,” *arXiv preprint arXiv:2502.15442*, 2025.
- [22] J. A. Collins, C. Houff, Y. L. Tan, and C. C. Kemp, “ForceSight: Text-guided mobile manipulation with visual-force goals,” in *ICRA*, 2024, arXiv:2309.12312.
- [23] M. Shukor *et al.*, “SmolVLA: A vision-language-action model for affordable and efficient robotics,” *arXiv preprint arXiv:2506.01844*, 2025.
- [24] K. Black *et al.*, “ π_0 : A vision-language-action flow model for general robot control,” *arXiv preprint arXiv:2410.24164*, 2024.
- [25] S. Ross, G. J. Gordon, and J. A. Bagnell, “A reduction of imitation learning and structured prediction to no-regret online learning,” in *AISTATS*, 2011.